



**COMSATS University Islamabad,
Sahiwal Campus.**

CUI Internship Management System

By

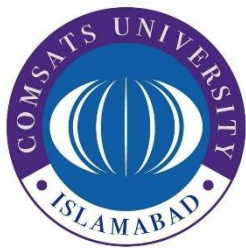
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Bachelor of Science in Computer Science (2022-2026)



**COMSATS University Islamabad,
Sahiwal Campus.**

CUI Internship Management System

A Project Presented To

**COMSATS University Islamabad,
Sahiwal Campus**

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Computer Science (2022-2026)

By

Muhammad Talha Tariq

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We hereby declare that this software, neither whole nor as a part has been copied out from any source. It is further declared that we have developed this software and accompanied report entirely based on our personal efforts. If any part of this project is proved to be copied out from any source or found to be reproduction of some other. We will stand by the consequences. No Portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

M Zain Ul Abideen

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CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS(SE) “CUI Internship Management System” was developed by **Muhammad Talha Tariq (CIIT/FA22-BCS-083/SWL)** and **Muhammad Zain ul Abideen (CIIT/FA22-BCS-090/SWL)** under the supervision of “**Ms. Raheela Shahzadi**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Science.

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Executive Summary

The Internship Management System is a dedicated digital platform designed to streamline and manage the complete internship lifecycle for university students, administrators, supervisors, and internship providers. Unlike traditional manual processes, this system provides a centralized solution for managing internship activities efficiently, improving communication, transparency, and performance tracking throughout the internship period.

The platform supports four major user roles: students, administrators, university supervisors, and internship providers (sites). Students can manage internship records, submit weekly progress reports, and track their evaluations and deadlines. Administrators can oversee the complete internship workflow, manage student placements, assign supervisors, and maintain internship records. University supervisors can monitor assigned student groups, review progress reports, and evaluate performance. Internship providers can assess student performance through structured evaluation forms provided by the university.

A key feature of the system is its structured internship monitoring mechanism, where students are grouped according to their internship sites and supervisors are assigned to monitor multiple groups. This ensures organized supervision, regular progress tracking, and efficient evaluation management. The system also calculates final student performance based on multiple evaluation components, including site feedback, supervisor assessment, and presentation marks.

To improve user support and system accessibility, an AI-powered chatbot is integrated into the platform to provide instant guidance, answer internship-related queries, assist with report submissions, and provide real-time information about deadlines, supervisors, and grading criteria.

The development of the system follows an agile methodology, allowing features to be developed incrementally and refined through continuous feedback. Functional requirements define the core workflow of the platform, while non-functional requirements focus on security, usability, reliability, and scalability.

Testing and evaluation include unit testing, integration testing, system testing, and user acceptance testing to ensure the system performs reliably and meets stakeholder requirements.

In summary, the Internship Management System modernizes the internship management process by automating administrative tasks, improving communication between stakeholders, enhancing supervision efficiency, and providing intelligent assistance through AI integration. This system serves as a comprehensive solution to improve internship management and student performance evaluation within the university environment.

Acknowledgment

All praise is to almighty Allah who bestowed upon us a minute portion of his boundless knowledge by which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor “**Ms. Raheela Shahzadi**”. Without their supervision, advice, and valuable guidance, the completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

M Zain Ul Abideen

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Abbreviations

API	Application Programming Interface
JS	JavaScript
SRS	Software Require Specification
SDD	Software Design Description
GUI	Graphical User Interface
IDEs	Integrated development environments
CSS	Cascading Style Sheet

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Chapter 1

Introduction

1 Introduction

The Internship Management System is a digital platform developed to address the challenges associated with managing internship activities for students at COMSATS University Islamabad. Internships are an essential part of academic learning, providing students with practical industry experience and professional exposure. However, the traditional internship management process often involves manual record keeping, fragmented communication, and inefficient progress monitoring, which creates difficulties for students, university staff, supervisors, and internship providers.

This system is specifically designed to streamline and automate the internship process by providing a centralized platform for all stakeholders involved. It supports four major user roles: students, administrators, university supervisors, and internship providers (sites). Each role is equipped with dedicated functionalities to ensure smooth communication, efficient supervision, and transparent evaluation throughout the internship period.

Students can use the platform to manage their internship information, submit weekly progress reports, and monitor deadlines and evaluation progress. Administrators are responsible for managing student placements, assigning supervisors, organizing students into internship groups, and overseeing the complete internship workflow. University supervisors can monitor assigned groups, review weekly reports, and evaluate student performance, while internship providers can submit structured performance evaluations directly through the system.

Currently, there is no dedicated digital platform at COMSATS University Islamabad for managing the complete internship lifecycle in an organized and efficient manner. Most processes are handled manually, which often leads to communication gaps, delays in evaluation, and difficulty in tracking student progress. The Internship Management System aims to solve these problems by offering a comprehensive and reliable solution for internship coordination and management.

An important feature of the system is the integration of an AI-powered chatbot, designed to assist users by answering internship-related queries, guiding students regarding report submissions, providing deadline information, and offering quick access to supervisor and grading details. This intelligent support improves accessibility and reduces dependency on manual guidance.

The system is designed with a user-friendly interface to ensure ease of use for all stakeholders, regardless of their technical background. Its intuitive design simplifies tasks such as report submission, evaluation management, and progress monitoring, making the platform accessible and efficient for all users.

In essence, the Internship Management System addresses the operational and communication challenges of internship management by providing an organized, transparent, and intelligent platform. It improves coordination among all stakeholders and enhances the overall internship experience for students and university administration.

1.1 Brief Overview

The Internship Management System project aims to develop a centralized web and mobile-based application for COMSATS University Islamabad to efficiently manage and monitor the complete internship process for students. The system is designed to digitalize internship operations, improve communication among stakeholders, and automate progress tracking and evaluation procedures.

The platform is being developed using modern web technologies, with Angular for the frontend and Next.js for the backend, ensuring scalability, performance, and maintainability. The development process follows the agile software development methodology, allowing iterative implementation, continuous improvements, and adaptability to changing requirements.

The Internship Management System is specifically designed to handle the structured workflow of university internships by connecting students, administrators, university supervisors, and internship providers within a single platform. Students can submit weekly reports, manage internship details, and monitor their progress. Administrators can manage internship placements, assign supervisors, and oversee all internship activities. Supervisors can evaluate students and review their weekly progress, while internship providers can submit performance evaluations directly into the system.

A unique aspect of this system is its structured supervision model, where students are grouped according to internship sites, and supervisors are assigned to monitor these groups. This organized approach improves internship monitoring and ensures efficient communication between all involved parties.

Overall, the Internship Management System provides a practical and efficient solution for modernizing internship management at the university by reducing manual work, improving transparency, and enhancing the internship experience for both students and administrators.

1.2 Relevance to Course Modules

The development of the Internship Management System directly applies and integrates various concepts, techniques, and practical skills acquired throughout the Computer Science degree program, particularly from software engineering and related technical courses. The project reflects the practical implementation of academic knowledge into a real-world system designed to solve organizational and communication challenges in internship management.

i. Technical Skills

The project applies programming languages, frameworks, and development tools learned during academic coursework, particularly in full-stack web development. Technologies such as Angular for frontend development and Next.js for backend services are used to build a scalable and responsive system. API development, authentication mechanisms, and frontend-backend integration are key technical implementations in this project.

ii. Software Engineering Principles

Adherence to software engineering principles, encompassing software design, development methodologies, testing procedures, and project management. This subject influenced requirement analysis, system design, coding, testing, and maintenance phases, ensuring a comprehensive software engineering perspective [1]

iii. Database Design

Database management knowledge plays a crucial role in designing the system's data structure. The project requires efficient database schema design for managing students, internship sites, supervisors, weekly reports, evaluations, and grading records. Concepts such as normalization, indexing, and relational mapping are applied to ensure data integrity and efficient retrieval.

iv. System Architecture

Understanding software architecture and design patterns is essential for building a modular, maintainable, and scalable system. The project involves designing separate modules for different user roles and ensuring smooth communication between frontend, backend, and database layers while maintaining clear separation of concerns.

v. UI/UX Design

A user-friendly interface is critical for ensuring effective user interaction. Concepts from human-computer interaction and UI/UX design courses are applied to create intuitive dashboards, report submission workflows, and management interfaces for students, supervisors, administrators, and internship providers.

vi. Project Management:

Project management concepts contribute to planning, scheduling, and controlling the development process. Agile project management practices support requirement prioritization, task division, milestone tracking, and timely feature delivery.

vii. Networking

The system depends on networking concepts for API communication, authentication flows, data exchange, and secure client-server interaction. Understanding HTTP protocols, REST APIs, and secure communication mechanisms is essential for system functionality.

1.3 Project Background

The need for the Internship Management System arises from the operational challenges and inefficiencies observed in the traditional internship management process at COMSATS University Islamabad. Internship management involves coordination between multiple stakeholders, including students, administrators, university supervisors, and internship providers, but the absence of a centralized digital system creates several management and communication issues.

Prior to this project, most internship-related activities were handled manually or through scattered communication channels, which often resulted in delays, miscommunication, and poor record management. Students faced difficulties in tracking deadlines, submitting weekly reports, and staying updated about internship requirements. Supervisors experienced challenges in monitoring student progress efficiently, especially when managing multiple student groups across different internship sites.

Additionally, administrators faced difficulties in organizing student placements, assigning supervisors, maintaining internship records, and collecting evaluations from internship providers. The lack of a structured system made it difficult to maintain transparency and monitor the complete internship workflow effectively.

Another major challenge was the evaluation process of final internship reports. Manual assessment of reports required significant time and effort from supervisors, and maintaining consistency in evaluation standards was difficult. This highlighted the need for an AI-supported evaluation mechanism that could assist in analyzing report quality, content completeness, and relevance.

In response to these challenges, the Internship Management System project was conceived as a centralized solution to automate and streamline the internship lifecycle. The system provides structured report submission, organized supervision, digital performance evaluation, and AI-assisted final report assessment.

By addressing these limitations, the project aims to improve efficiency, strengthen communication between stakeholders, enhance transparency in evaluation, and modernize the overall internship management process within the university.

1.4 Methodology and Software Lifecycle

The Internship Management System is a multi-stakeholder project with complex functional requirements, including student management, supervisor coordination, report submission, evaluation handling, and AI-based report assessment. Due to evolving requirements and the need for continuous improvement, a flexible and iterative development approach is required.

The Agile software development model has been selected for this project as it supports adaptability [2], collaboration, and incremental development. It allows continuous feedback from students, administrators, supervisors, and internship providers, ensuring the system aligns with real operational needs.

Agile enables the system to be developed in modular increments, making it easier to manage complexity and accommodate requirement changes. Key components such as internship management, supervisor monitoring, weekly report submission, performance evaluation, and AI-based final report analysis are developed and refined progressively.

1.4.1 Rationale behind Selected Agile Model

Agile is a software development methodology that emphasizes adaptability, collaboration, and continuous delivery. It is highly suitable for the Internship Management System because project

requirements can evolve during development as stakeholders identify new operational needs or improvements.

The involvement of different stakeholders throughout the system development process ensures that feedback is continuously collected and incorporated into future iterations. This helps in improving usability, fixing operational gaps, and enhancing system efficiency.

Agile is based on the following principles:

Iterative development: The system is developed in multiple iterations, where each cycle introduces new modules or improves existing functionalities such as report submission, supervisor dashboards, and evaluation systems.

Incremental delivery: Functional parts of the system are delivered step by step, allowing stakeholders to review and use complete modules while development continues.

Continuous feedback: Students, administrators, supervisors, and internship providers provide regular feedback, helping improve workflows and user experience.

Adaptive planning: Development planning remains flexible, allowing modifications based on stakeholder requirements, operational challenges, and system improvement needs.

Chapter 2

Problem Definition

2 Problem Definition

This chapter defines the core problem addressed by the project and outlines the intended outcomes of the proposed solution.

2.1 Problem Statement

The Internship Management System is designed to address the inefficiencies and lack of centralization in the current internship management process at COMSATS University Islamabad. At present, internship-related activities are managed through manual processes and fragmented communication channels, which leads to difficulties in coordination among students, administrators, university supervisors, and internship providers.

Students face challenges in tracking internship requirements, submitting weekly reports, and staying updated with deadlines and evaluation criteria. Supervisors struggle to efficiently monitor multiple student groups and maintain consistent evaluation standards. Administrators encounter difficulties in managing student placements, assigning supervisors, collecting performance feedback, and maintaining accurate internship records. Internship providers also lack a structured and efficient system for submitting evaluations and communicating with the university.

Another significant issue is the absence of an automated evaluation mechanism for final internship reports. Manual assessment is time-consuming, prone to inconsistency, and difficult to standardize across different supervisors. This creates a need for a more structured and intelligent evaluation approach.

Currently, there is no unified digital platform within the university that integrates all internship management functions, including report submission, supervision, evaluation tracking, and AI-assisted assessment. The absence of such a system highlights a critical gap in the existing academic and administrative infrastructure.

The Internship Management System aims to resolve these issues by providing a centralized, role-based platform that streamlines internship operations, improves communication, enhances transparency, and introduces AI-based support for final report evaluation.

2.2 Deliverables and Development Requirements

The Internship Management System includes a set of deliverables and development requirements necessary to ensure successful implementation, deployment, and long-term maintainability of the platform

- **User Interface (UI) Design:** Develop an intuitive and role-based interface for students, administrators, supervisors, and internship providers. The design should support ease of navigation for tasks such as report submission, evaluation review, internship tracking, and dashboard monitoring.
- **Internship Workflow Management:** Implement structured workflows for managing student internships, including placement assignment, group formation based on internship sites, supervisor allocation, and tracking of internship progress.
- **Report Submission System:** Design a system that allows students to submit weekly progress reports and final internship reports in a structured format, ensuring proper storage and retrieval.
- **Evaluation and Grading Module:** Develop a module for supervisors and internship providers to evaluate student performance. This includes site feedback, supervisor assessment, presentation evaluation, and automated grading support.
- **AI-Based Report Evaluation:** Integrate an AI-powered module to assist in analyzing final internship reports by assessing content quality, completeness, and relevance to support supervisors in evaluation decisions.
- **Authentication and Role Management:** Implement a secure authentication system with role-based access control for students, administrators, supervisors, and internship providers.
- **Scalability:** Design the system architecture to support a growing number of users, internship records, and concurrent operations without performance degradation.
- **Security and Data Protection:** Ensure protection of sensitive academic and personal data through secure authentication mechanisms, encrypted communication, and controlled access to system resources.
- **Cross-Platform Accessibility:** Develop the system as a web-based application to ensure accessibility across multiple devices and platforms without requiring platform-specific installations.
- **Documentation:** Provide comprehensive technical documentation and user guides for system usage, maintenance, and future enhancements.

- **Testing and Quality Assurance:** Conduct unit testing, integration testing, system testing, and user acceptance testing to ensure system reliability, accuracy, and performance.
- **Regular Updates and Maintenance:** Plan for continuous maintenance and future enhancements to accommodate evolving academic requirements and potential feature expansions.

These deliverables collectively ensure the successful development of the Internship Management System, aiming to improve internship coordination, enhance evaluation accuracy, and provide a centralized platform for all stakeholders involved.

2.3 Current System

At present, COMSATS University Islamabad does not have a centralized digital system for managing the complete internship lifecycle. Internship-related activities are largely handled through manual processes, spreadsheets, email communication, and informal coordination between students, supervisors, administrators, and internship providers.

This fragmented approach creates several operational challenges. Students often face difficulties in tracking internship requirements, submitting weekly reports on time, and accessing clear evaluation criteria. Communication with supervisors is inconsistent, which can lead to delays in feedback and unclear guidance regarding performance expectations.

Administrators manage internship placements, supervisor assignments, and record maintenance manually, which increases workload and introduces the risk of data inconsistency and errors. Similarly, supervisors must monitor multiple student groups without a structured dashboard, making it difficult to efficiently review progress and maintain uniform evaluation standards.

Internship providers also lack a streamlined method for submitting structured feedback and evaluations, resulting in delayed or incomplete assessment data being shared with the university.

In addition, the absence of an automated or AI-assisted evaluation mechanism for final internship reports further increases the manual burden on supervisors and reduces consistency in grading outcomes..

Chapter 3

Requirement Analysis

3 Requirement Analysis

3.1 Use Case Diagrams

In figure 3.1, we show the behavior of our system. This enable you to visualize the different types of roles in a system and how those roles interact with the system.

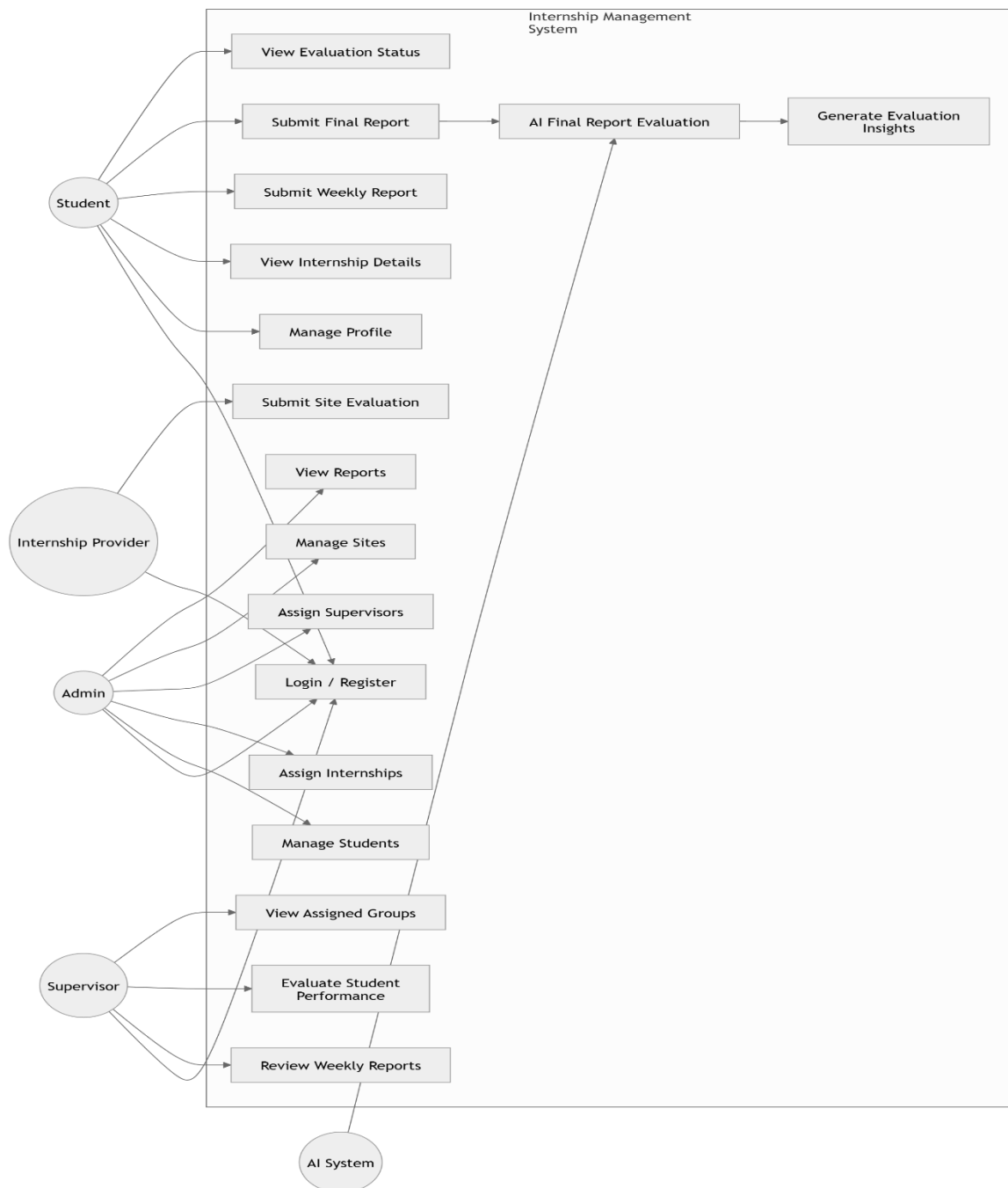


Figure 3.1 CUI Internship Management System Use Case

3.2 Use Case Descriptions

The table below indicate comprehensive use case descriptions

Table 3.1 Use case description for login/logout

Field	Description
Use Case ID	UC-01
Use Case Name	Login / Logout
Actors	Primary Actor: Student, Admin, Supervisor, Internship Provider
Description	This use case defines the authentication process for all system users. It allows users to securely log in to access the system and log out after completing their session.
Trigger	User wants to access the system.
Preconditions	User must be registered in the system.

Table 3.2 Use case description for user registration

Field	Description
Use Case ID	UC-02
Use Case Name	Register User
Actors	Primary Actor: Student
Description	This use case allows new students to register in the system using valid university credentials before accessing internship services.
Trigger	User initiates registration process.
Preconditions	User must be a valid student of COMSATS University Islamabad with a valid email.

Table 3.3 Use case description for *Manage Internship Profile*

Field	Description
Use Case ID	UC-03

Use Case Name	Manage Internship Profile
Actors	Primary Actor: Student
Description	Students can view and manage their internship details, including assigned site, supervisor, and progress status.
Trigger	User selects profile management option.
Preconditions	User must be registered and logged in.

Table 3.4 Use case description for *Submit Weekly Report*

Field	Description
Use Case ID	UC-04
Use Case Name	Submit Weekly Report
Actors	Primary Actor: Student
Description	Students submit weekly progress reports describing their internship activities and learning outcomes.
Trigger	Student chooses to submit weekly report.
Preconditions	Student must be assigned to an internship.

Table 3.5 Use case description for *Submit Final Report*

Field	Description
Use Case ID	UC-05
Use Case Name	Submit Final Report
Actors	Primary Actor: Student
Description	Students submit their final internship report for evaluation at the end of the internship period.
Trigger	Student completes internship and submits final report.
Preconditions	All weekly reports must be completed or available.

Table 3.6 Use case description for *Evaluate Student Performance*

Field	Description
Use Case ID	UC-06
Use Case Name	Evaluate Student Performance
Actors	Primary Actor: Supervisor
Description	Supervisors evaluate student performance based on weekly reports, behavior, and overall internship progress.
Trigger	Supervisor reviews assigned student group.
Preconditions	Students must be assigned to supervisor.

Table 3.7 Use case description for *Manage Students*

Field	Description
Use Case ID	UC-07
Use Case Name	Manage Students
Actors	Primary Actor: Admin
Description	Administrators manage student records, internship assignments, and group formations.
Trigger	Admin accesses student management module.
Preconditions	Admin must be authenticated in the system.

Table 3.8 Use case description for deleting *AI Final Report Evaluation*

Field	Description
Use Case ID	UC-08
Use Case Name	AI Final Report Evaluation
Actors	Primary Actor: AI System, Supervisor
Description	The system uses AI to analyze final internship reports and generate evaluation insights to assist supervisors in grading.
Trigger	Final report is submitted by student.
Preconditions	Final report must be available in the system.

3.3 Functional Requirements

Functional requirements define the core features and operations that the system must provide to enable users to perform their tasks effectively [3]. The Internship Management System is composed of several modules designed to support students, administrators, supervisors, and internship providers in managing the complete internship lifecycle.

This system will have the following modules:

3.3.1 User Registration

Users must be able to register in the system using a valid university email address issued by COMSATS University Islamabad. The email address serves as the unique identifier for each user account in the system.

3.3.2 Login / Authentication

Registered users must be able to securely log in to the system using their credentials. The system should validate user identity and provide role-based access (Student, Admin, Supervisor, Internship Provider).

3.3.3 Internship Assignment

Administrators should be able to assign students to internship sites and allocate supervisors to each student group based on predefined rules and availability.

3.3.4 Weekly Report Submission

Students must be able to submit weekly progress reports describing their internship activities. These reports should be accessible to assigned supervisors for review.

3.3.5 Final Report Submission

Students should be able to submit a final internship report at the completion of their internship for evaluation purposes.

3.3.6 Evaluation Management

Supervisors and internship providers must be able to evaluate student performance based on weekly reports, behavior, and overall internship performance.

3.3.7 AI-Based Report Evaluation

The system should support AI-assisted analysis of final internship reports to generate evaluation insights and assist supervisors in grading consistency.

3.3.8 Student Profile Management

Students should be able to view and update their internship-related profiles, including personal information, assigned site, and supervisor details.

3.4 Nonfunctional Requirements

Non-functional requirements define the quality attributes and constraints of the system, ensuring its performance, reliability, and usability under real-world conditions.

3.4.1 Performance

The system should efficiently handle multiple concurrent users, ensuring fast response times for operations such as report submission, data retrieval, and evaluation processing.

3.4.2 Security

Strong security mechanisms must be implemented to protect sensitive academic data, user credentials, and evaluation records. Role-based access control and secure authentication are required.

3.4.3 Availability

The system should be highly available and accessible 24/7 to ensure uninterrupted access for students, supervisors, administrators, and internship providers.

3.4.4 Ease of Use

The system should provide a simple, intuitive, and user-friendly interface to ensure ease of navigation for all users, regardless of technical expertise.

Chapter 4

Design and Architecture

4 Design and Architecture

4.1.1 System Architecture

The figure 4.1 explains the working of the system and describes the components of the CUI internship Management System, their relationships, and how they interact with each other, with this architecture [4].

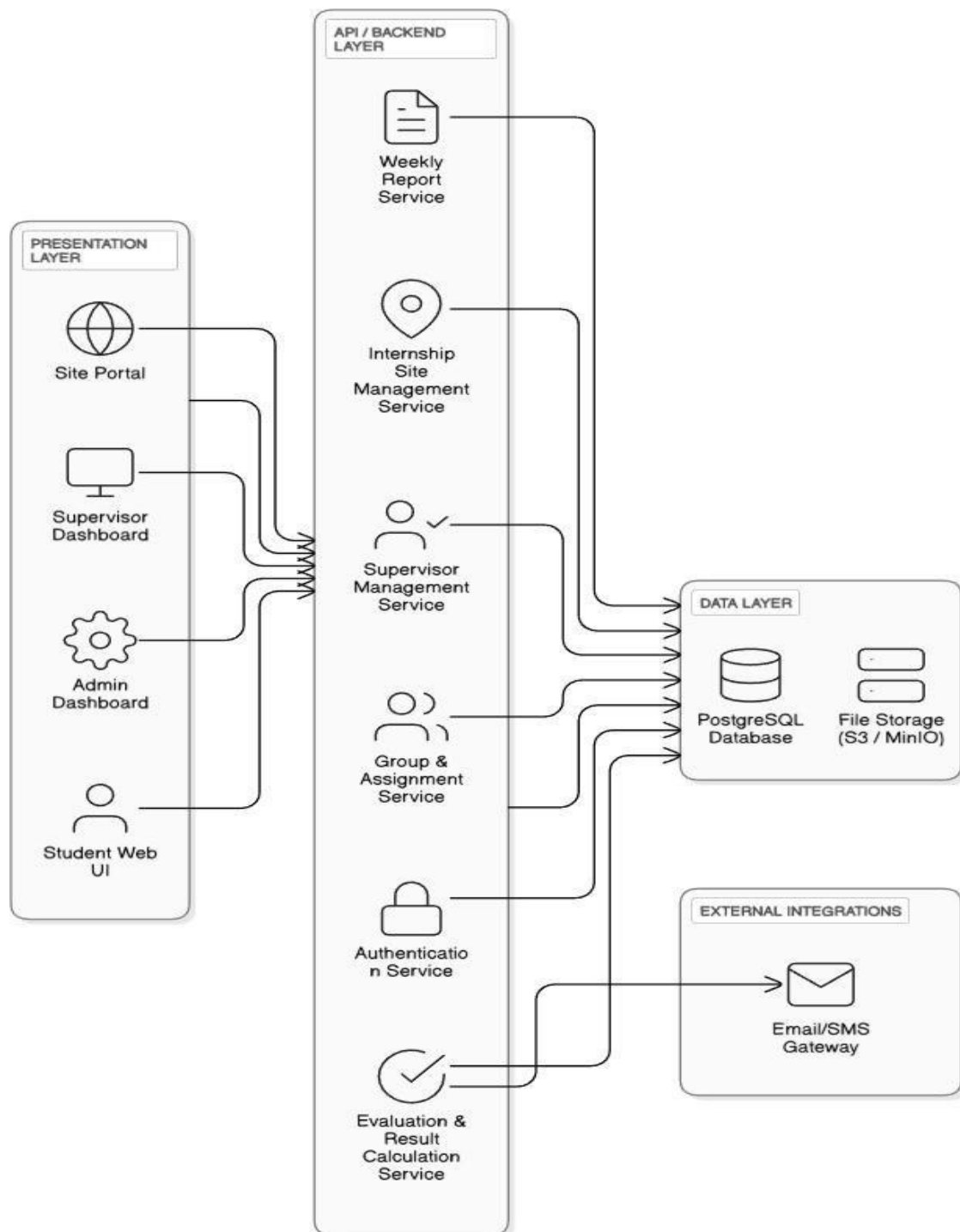


Figure 4.1 System architecture

4.2 Process Flow/Representation

The figure 4.2 Process Flow Representation which illustrates the step-by-step sequence of actions and interactions between different components in the CUI internship Management System.

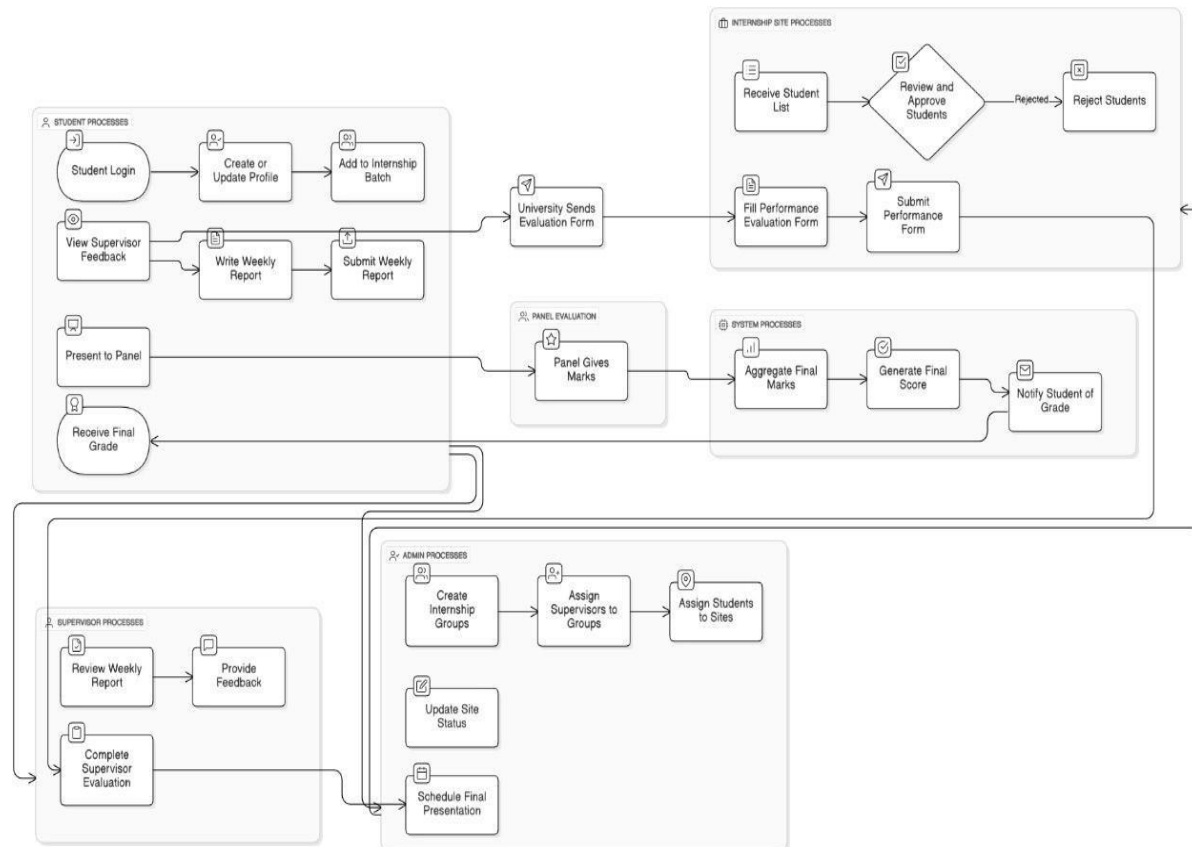


Figure 4.2 Process Flow

4.3 Sequence Diagram

These figures 4.3, 4.4, 4.5, 4.6, 4.7 and 4.8 show how the system will work through sequence diagrams, task after task sequence wise. It gives a complete description of the working of CUI internship Management System. [4]

User Authentication Module

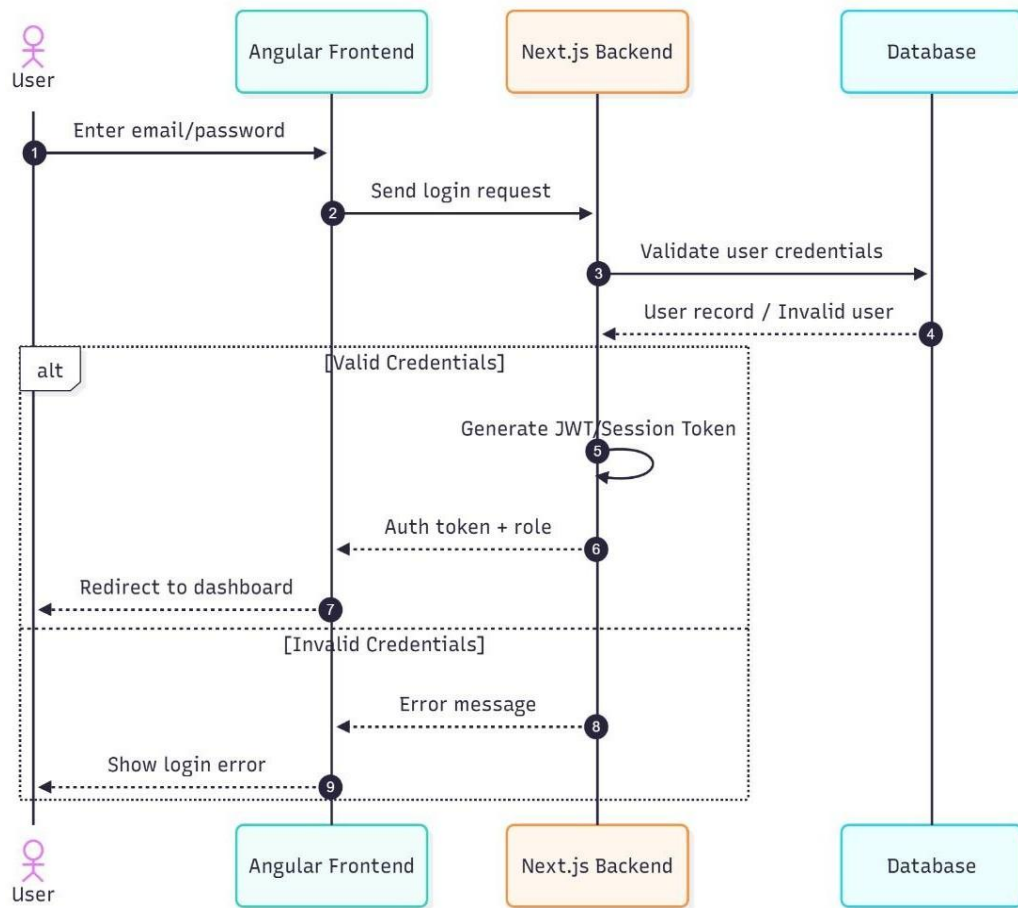


Figure 4.3 Authentication Sequence Diagram

Weekly Report Submission Module

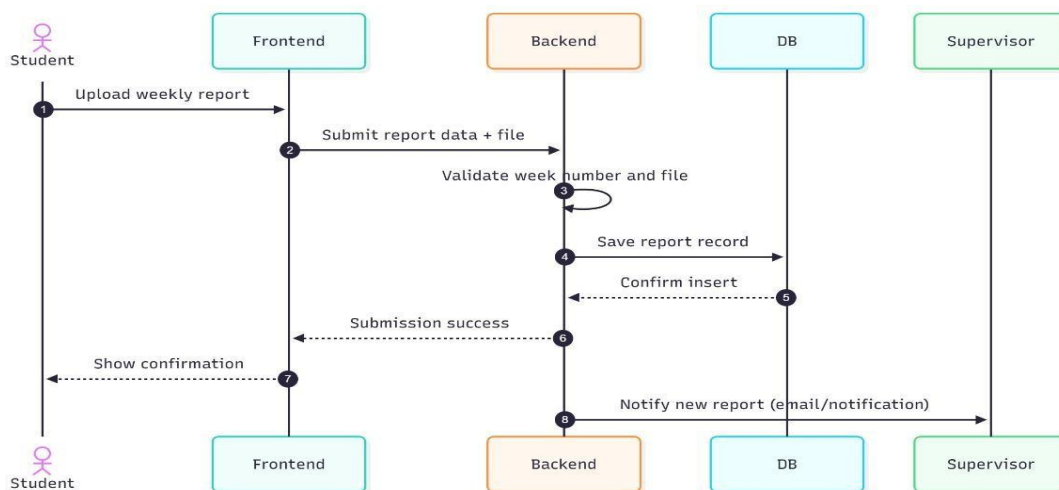


Figure 4.4 Weekly Report Sequence Diagram

Supervisor Review Module

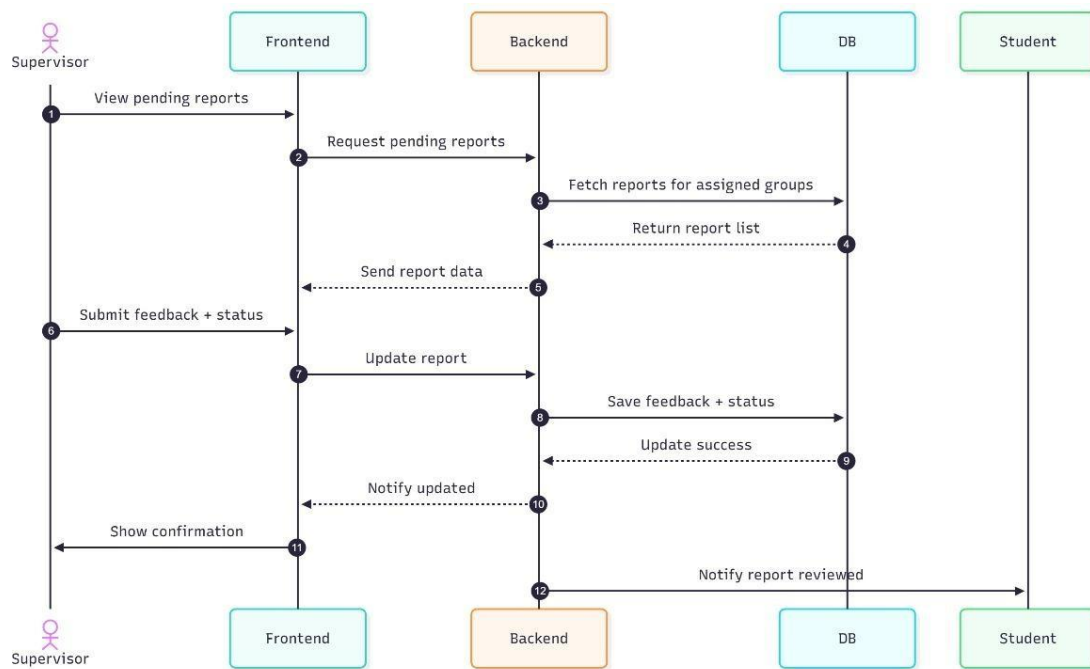


Figure 4.5 Supervisor Review Sequence Diagram

Site Performance Evaluation Module

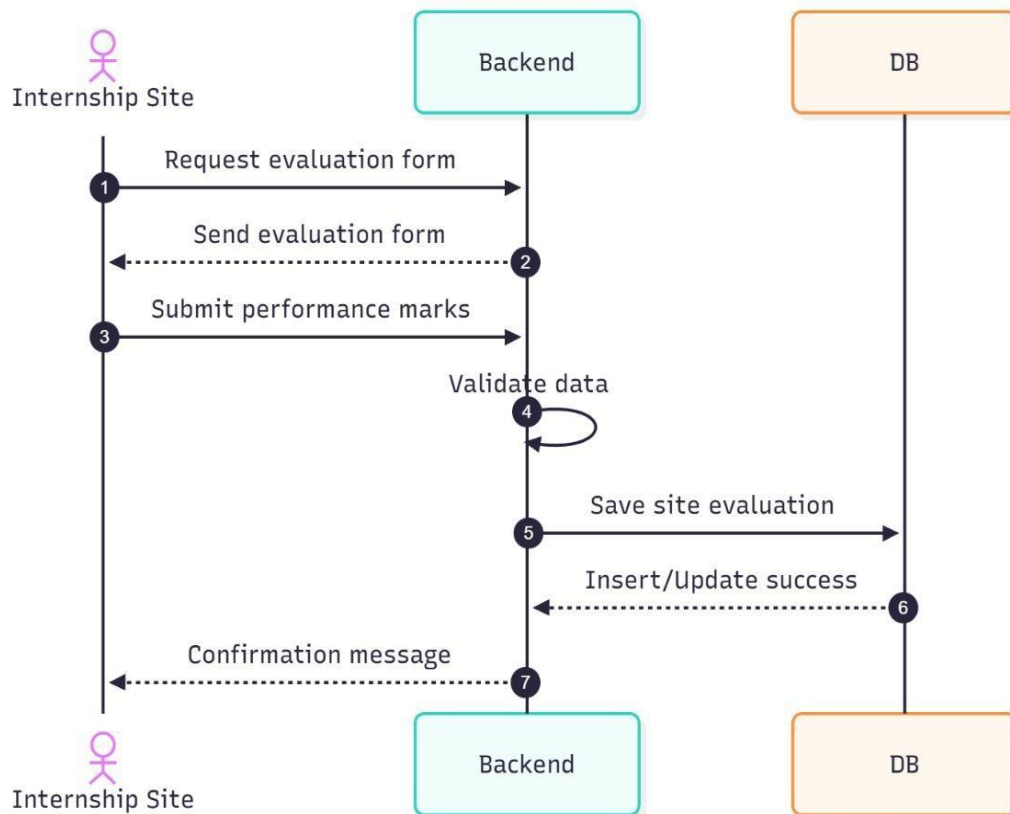


Figure 4.6 site Evaluation Sequence Diagram

Final Evaluation & Marks Calculation Module

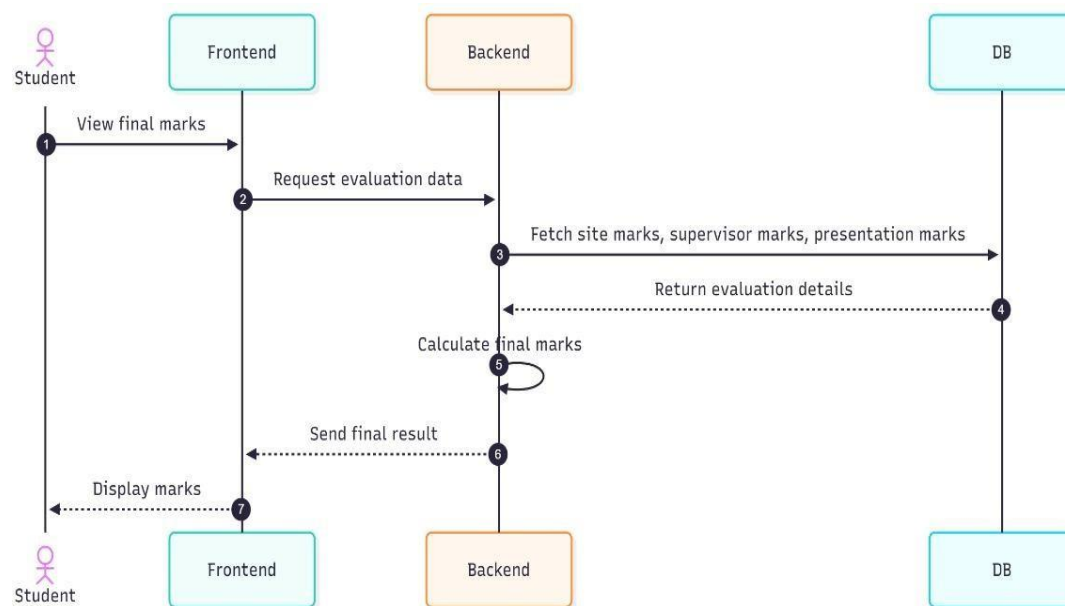


Figure 4.7 Final Evaluation Marks Sequence Diagram

Group Assignment Module (Admin)

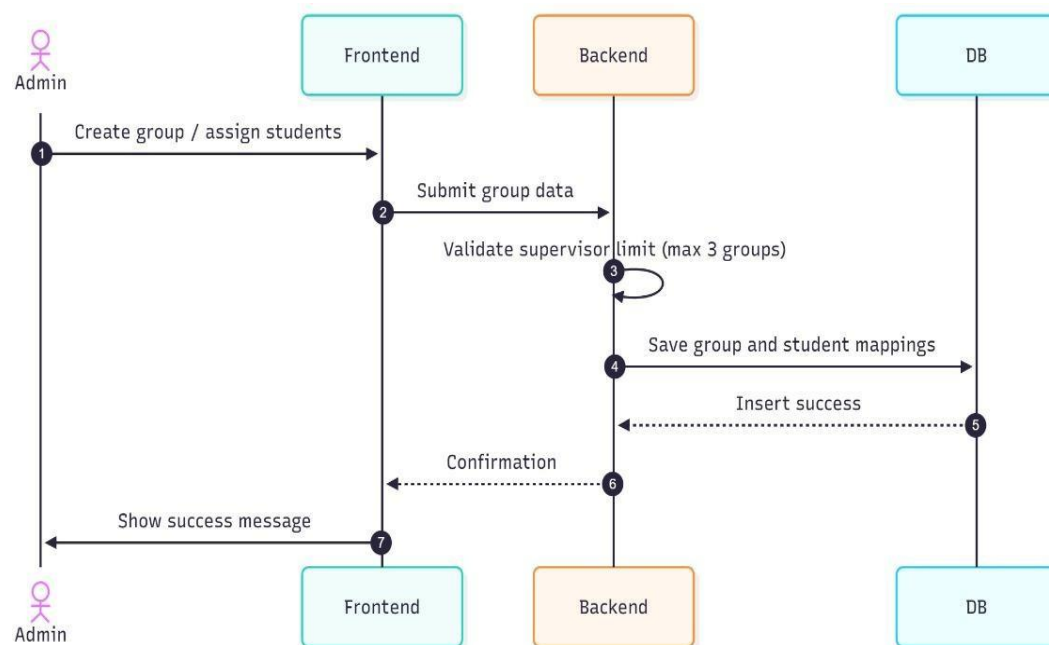


Figure 4.8 Group Assignment module Sequence Diagram

4.4 Class Diagram

The figure 4.9 shows class diagram which is the graphical representation of the system's classes, their attributes, and the relationships between them.

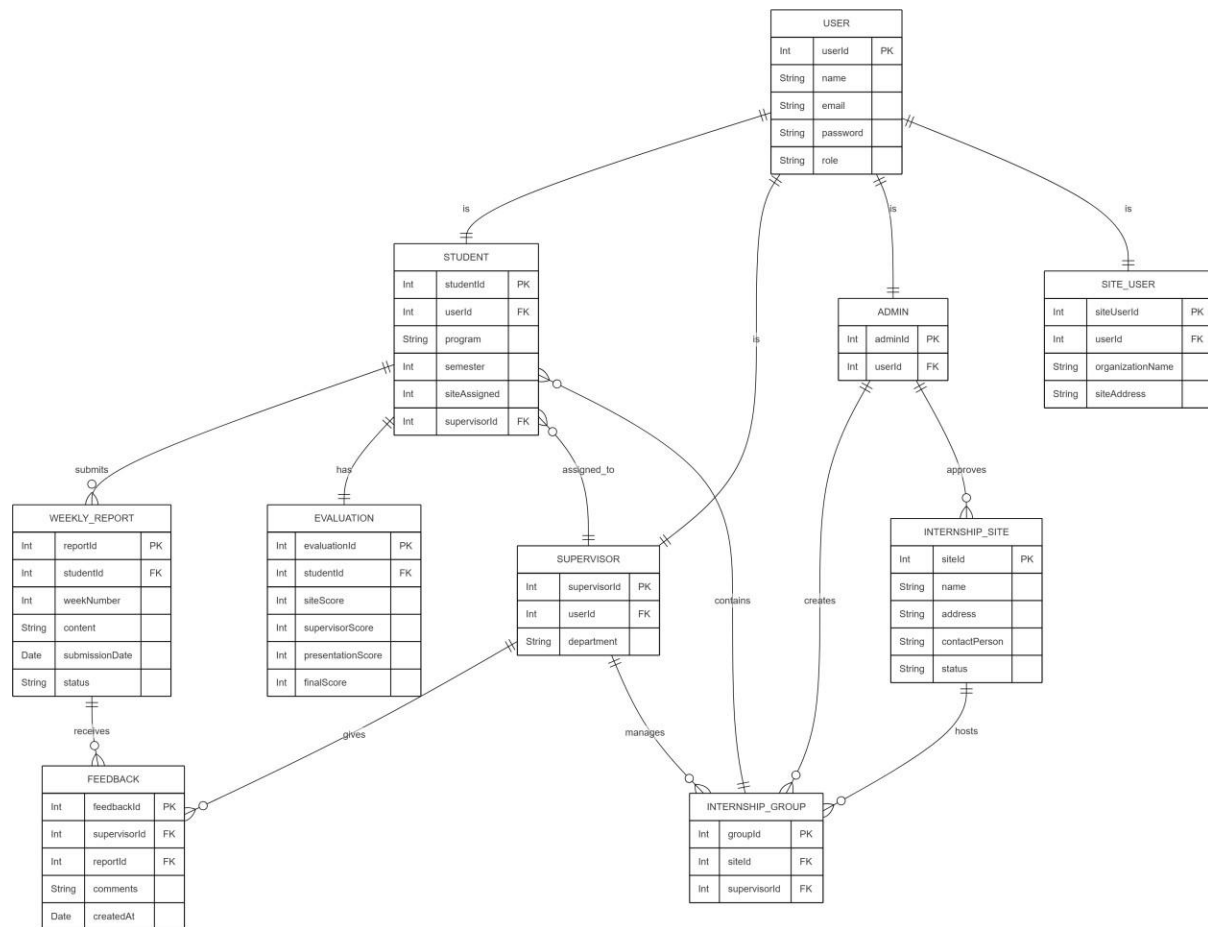


Figure 4.9 Class Diagram

Chapter 5

Implementation

5 Implementation

In the implementation phase of the Internship Management System, core algorithms and modules are developed to support smooth interaction between students, administrators, supervisors, and internship providers. The system focuses on automation of internship workflows, secure data handling, and AI-assisted evaluation of internship reports.

5.1 JWT-Based User Authentication

The system uses JSON Web Tokens (JWT) for secure authentication and authorization. After successful login, a token containing user identity and role information is generated [5]. This token is used for accessing protected routes and ensures secure role-based access control across the system. This approach improves scalability by eliminating the need for continuous session tracking.

5.2 User Registration Algorithm

This algorithm manages the registration process for new users (students, supervisors, administrators, and internship providers). It validates user credentials using university-issued email addresses and ensures secure storage of user data in the database. Role assignment is performed during registration or by the administrator depending on the system workflow.

5.3 Internship Workflow and Report Submission Algorithm

This algorithm governs the core internship operations, including student assignment, weekly report submission, and final report submission. Students submit weekly progress updates, which are stored in the system and made available to assigned supervisors. The workflow ensures proper tracking of internship progress throughout the internship duration.

5.4 AI-Based Final Report Evaluation Algorithm

A key component of the system is AI-assisted evaluation of final internship reports. The algorithm analyzes submitted reports using natural language processing techniques to assess content quality, completeness, and relevance [6]. It generates evaluation insights to assist supervisors in making consistent and efficient grading decisions.

5.5 User interface

The system provides a clean and structured user interface for all roles. The UI includes dashboards for students, administrators, supervisors, and internship providers, enabling easy navigation for tasks such as report submission, evaluation review, and internship tracking.

Home Screen

This figure 5.1 shows the landing page for the CUI Internship System, designed as a unified portal for students, faculty, and supervisors

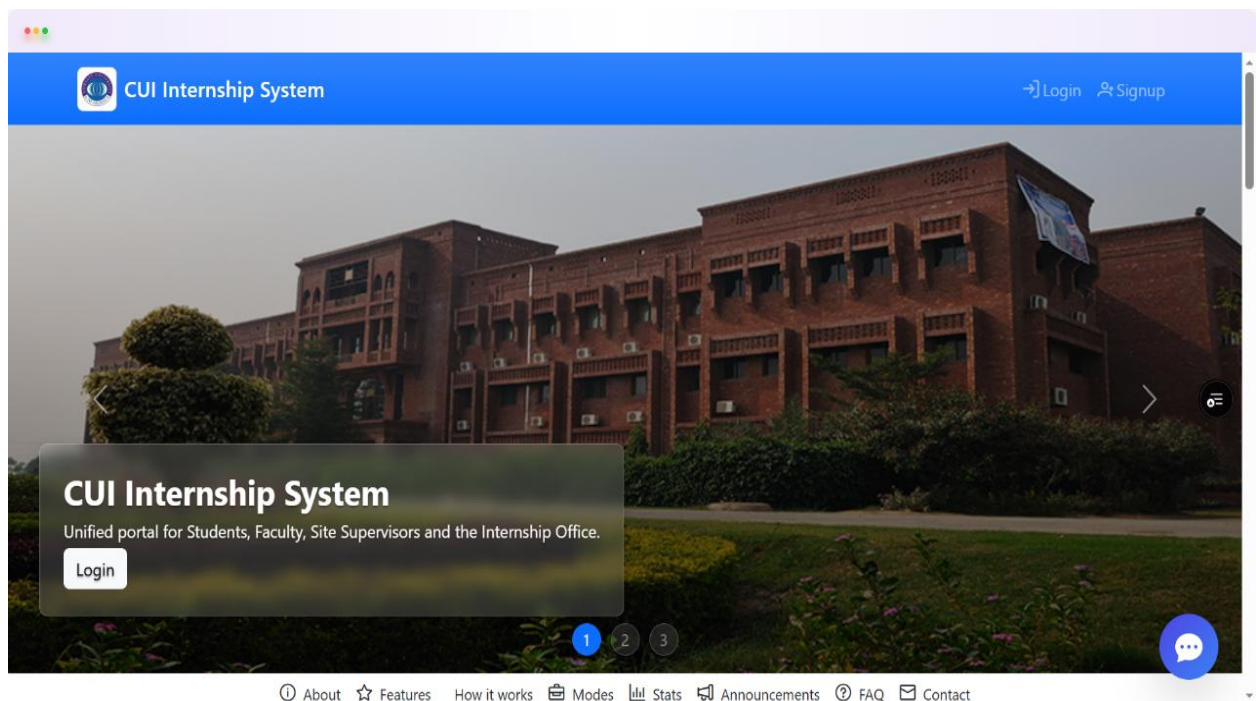
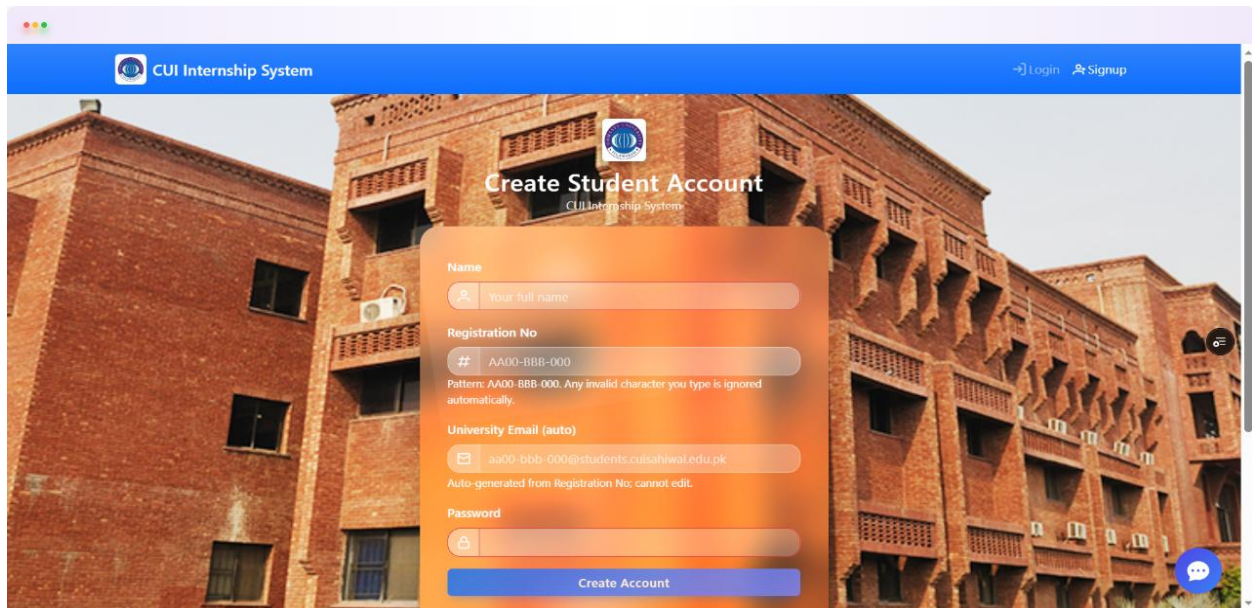


Figure 5.1 Home Screen

Signup Screen

This figure 5.2 shows a "Create Student Account" page for the CUI Internship System, featuring a registration form.



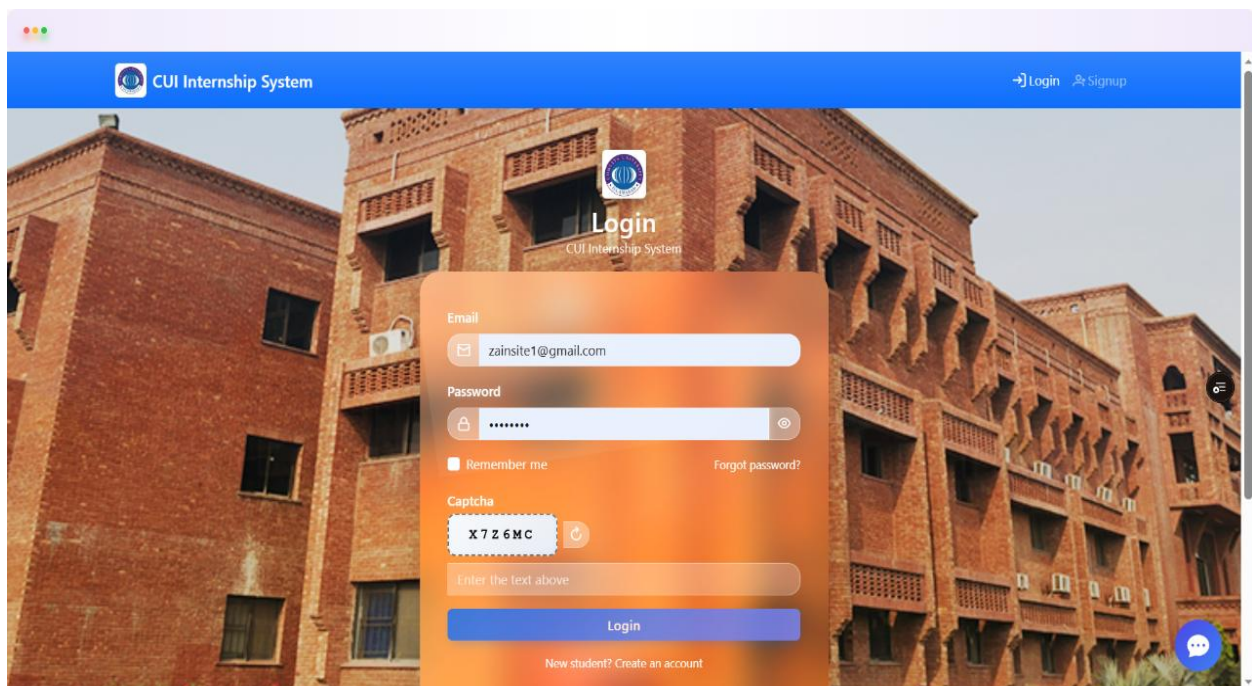
The screenshot displays the "Create Student Account" page for the CUI Internship System. The page has a blue header with the system name and navigation links for "Login" and "Signup". The main content area features a large background image of a red brick building. Overlaid on this is a registration form with the following fields and elements:

- Name:** A text input field with a placeholder "Your full name".
- Registration No:** A text input field with a placeholder "AA00-BBB-000". Below it, a note states: "Pattern: AA00-BBB-000. Any invalid character you type is ignored automatically."
- University Email (auto):** A text input field with a placeholder "aa00-bbb-000@students.cuisahiwal.edu.pk". Below it, a note states: "Auto-generated from Registration No; cannot edit."
- Password:** A text input field with a placeholder "Password".
- Create Account:** A blue button at the bottom of the form.

Figure 5.2 Sign Up Screen

Login Screen

This figure 5.3 shows a "Login " page for the CUI Internship System, featuring a login form.



The screenshot displays the "Login" page for the CUI Internship System. The page has a blue header with the system name and navigation links for "Login" and "Signup". The main content area features a large background image of a red brick building. Overlaid on this is a login form with the following fields and elements:

- Email:** A text input field with a placeholder "zainsite1@gmail.com".
- Password:** A text input field with a placeholder "Password".
- Remember me:** A checkbox labeled "Remember me".
- Forgot password?:** A link labeled "Forgot password?".
- Captcha:** A text input field with a placeholder "X7Z6MC".
- Login:** A blue button at the bottom of the form.
- New student? Create an account:** A link at the bottom of the form.

Figure 5.3 Login Screen

Student Dashboard

The figure 5.4 shows Student Dashboard which displays the details of an internship, including the assigned company, role, and duration.

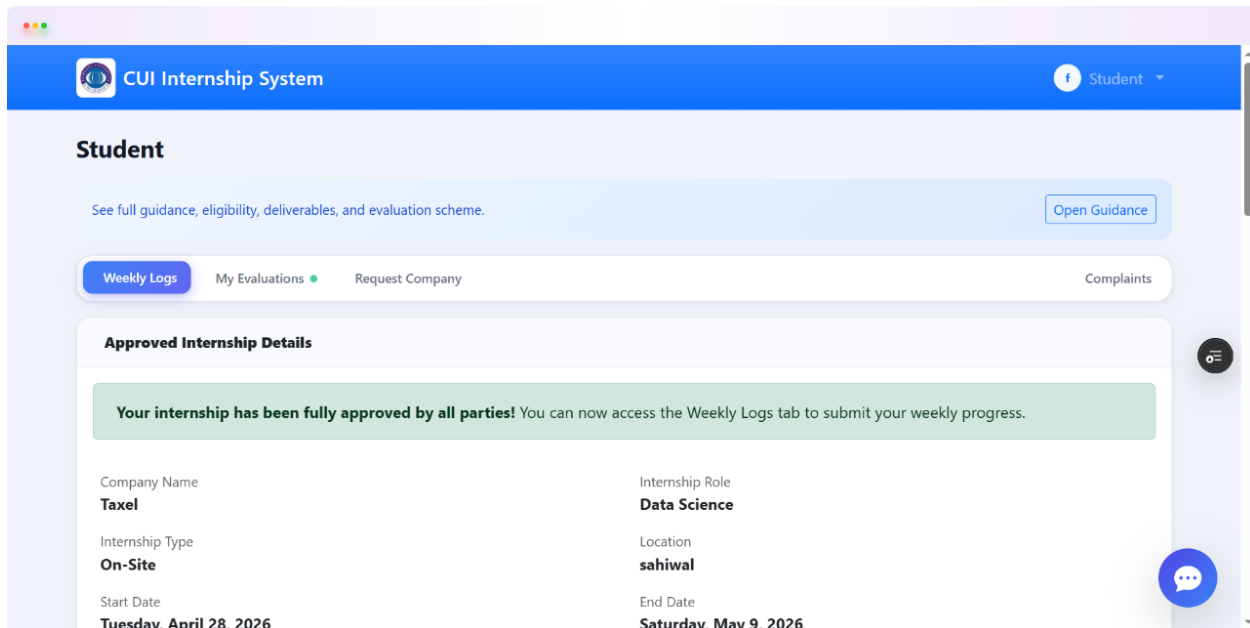


Figure 5.4 Student Dashboard

Final Result Screen

The figure 5.5 displays a **Final Result** dashboard that aggregates scores from various evaluators, including faculty and site supervisors, to provide an overall performance summary.

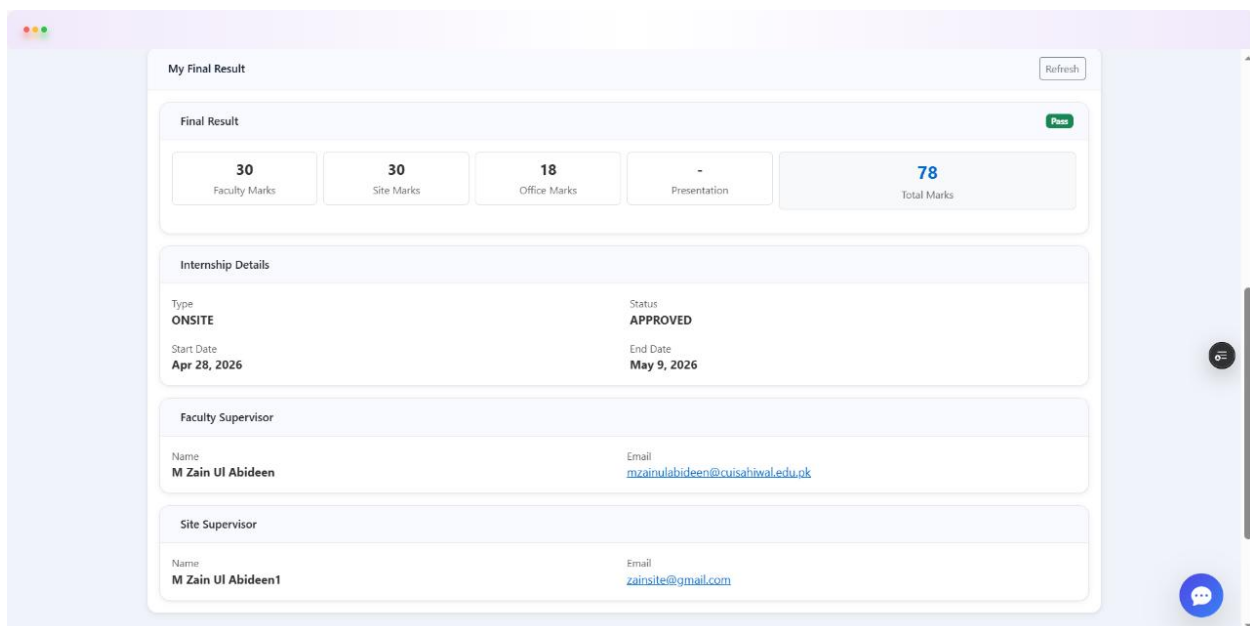


Figure 5.5 Student Final Result Screen

Internship Office Dashboard

The figure 5.5 displays the **Internship Office Dashboard**, specifically showing the **Announcements** management interface.

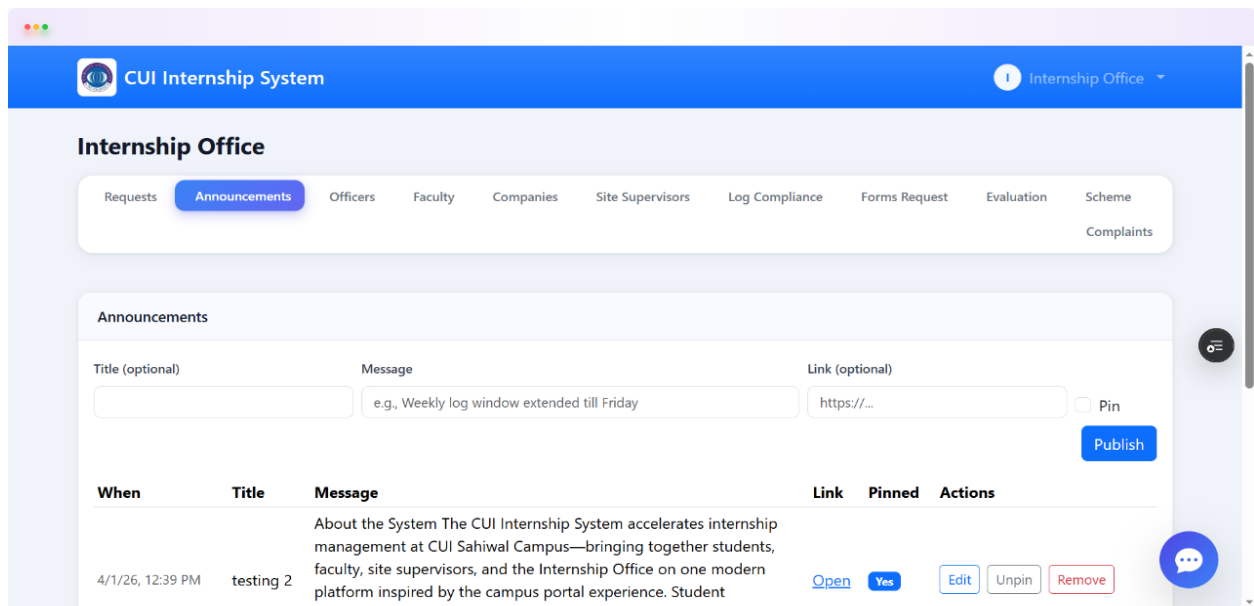


Figure 5.6 Admin Dashboard

Faculty Dashboard

The figure 5.7 shows **Faculty Supervisor Dashboard**, which provides a high-level overview of assigned students and their current approval statuses.

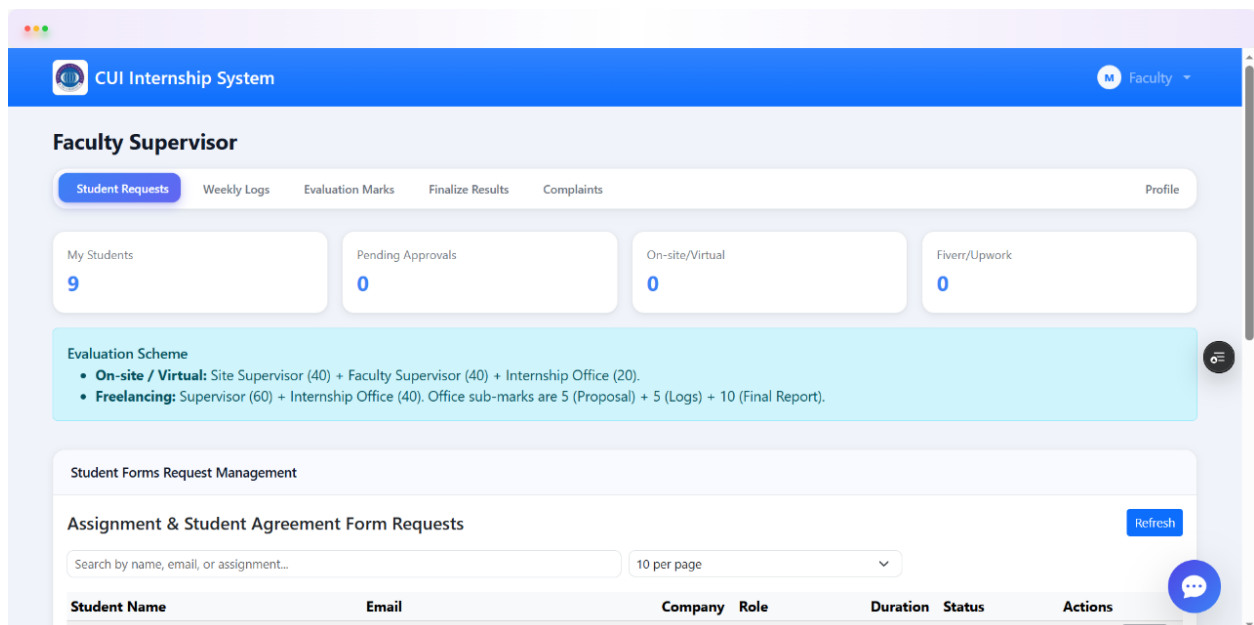


Figure 5.7 Faculty Dashboard

Site Supervisor Dashboard

The figure 5.8 features an "Assigned Students" list where external supervisors can manage interns hosted by their company. It includes status-based filtering (Pending, Approved, Completed, etc.) and quick-action buttons to view student details or input performance evaluations.

CUI Internship System

Site Supervisor

Students | Details | Evaluations

Assigned Students

Filter by Status

All | Pending | Approved | In Progress | Completed | Rejected

Showing 1 of 1 students

Name	Email	Reg No	Company	Status	Start Date	End Date	Actions
M Zain Ul Abideen	fa22-bcs-090@students.cuisahiwal.edu.pk	FA22-BCS-090	Taxel	APPROVED	3/11/26, 5:00 AM	4/11/26, 5:00 AM	Details Eval

Save All Scores

Figure 5.8 Site Dashboard

Chapter 6

Testing and Evaluation

6 Testing

6.1 Unit Testing

Unit testing was conducted to evaluate the individual components of the Internship Management System and ensure that each module performs as expected independently [7].

Table 6.1 Unit Testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
UT-001	User Registration	Valid user information	User account is created successfully	Pass
UT-002	User Authentication	Valid user credentials	Successful authentication and login	Pass
UT-003	Blank Password Field	Empty password during registration	System displays validation error	Pass
UT-004	Weekly Report Submission	Valid weekly report data	Report is submitted successfully	Pass
UT-005	Final Report Submission	Valid final report upload	Final report stored successfully	Pass
UT-006	AI Report Evaluation	Valid final report content	AI generates evaluation insights	Pass

6.2 Manual Testing

During the manual testing phase, the Internship Management System was thoroughly tested to verify system behavior, usability, and workflow accuracy. Identified issues were resolved to ensure system reliability and user satisfaction.

Table 6.2 Manual Testing

Test Cases	Expected Result	Actual Result	Pass/Fail
System Launch	Successful	Successful	Pass

User Registration	Successful	Successful	Pass
User Login	Successful	Successful	Pass
Dashboard Display	Accurate	Accurate	Pass
Internship Assignment	Successful	Successful	Pass
Weekly Report Submission	Successful	Successful	Pass
Final Report Submission	Successful	Successful	Pass
Supervisor Evaluation	Successful	Successful	Pass
AI Final Report Evaluation	Accurate	Accurate	Pass
Profile Management	Successful	Successful	Pass
System Navigation	Easy	Easy	Pass
System Performance	Smooth	Smooth	Pass
Logout	Successful	Successful	Pass

6.3 Functional Testing

Functional testing was performed to verify the complete system functionalities against the defined requirements and ensure that all modules behave correctly.

Table 6.3 Functional Testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
FT-001	User Registration	Valid university email registration	Account created successfully	Pass
FT-002	User Login	Valid credentials	User successfully authenticated	Pass
FT-003	Weekly Report Submission	Submitting internship progress report	Report successfully submitted	Pass
FT-004	Final Report Submission	Uploading final internship report	Report successfully uploaded	Pass
FT-005	Supervisor Evaluation	Evaluating student performance	Evaluation successfully stored	Pass
FT-006	AI Report Analysis	Final report content analysis	AI evaluation insights generated	Pass

6.4 Integration Testing

Integration testing was performed to validate the interaction between different modules of the Internship Management System, ensuring smooth communication and data flow.

Table 6.4 Integration testing

Test Case ID	Test Case Name	Attributes & Values Tested	Expected Result	Actual Result
IT-001	Frontend-Backend Integration	API communication between frontend and backend	Successful data exchange	Pass
IT-002	Backend-Database Integration	Database connectivity and data storage	Connection established successfully	Pass

Chapter 7

Conclusion and Future Work

7 Conclusion and Future Work

7.1 Conclusion

The Internship Management System represents a significant advancement in improving the internship management process at COMSATS University Islamabad. The development and implementation of this system address major challenges present in the traditional internship workflow, including manual record management, communication gaps, inefficient supervision, and inconsistent evaluation procedures.

The system provides a centralized and structured platform for students, administrators, university supervisors, and internship providers to manage the complete internship lifecycle efficiently. Through modules such as internship assignment, weekly report submission, final report submission, supervisor evaluation, and site performance feedback, the platform improves transparency, accountability, and workflow efficiency.

A major contribution of this project is the integration of AI-based final report evaluation, which assists supervisors in analyzing report quality, completeness, and relevance. This feature improves consistency in assessment and reduces the manual workload involved in report evaluation.

The user-friendly interface, structured workflows, secure authentication, and centralized data management make the system reliable and practical for academic use. The project successfully meets its objectives by providing an efficient, scalable, and intelligent solution for internship management while improving communication and coordination among all stakeholders.

7.2 Future Work

Although the initial implementation of the Internship Management System is successful, several future enhancements can further improve its functionality and effectiveness:

- **Advanced AI Evaluation:**

Enhancing the AI module to provide deeper report analysis, plagiarism detection, and more accurate performance insights,

- **Mobile Application Development:**

Developing dedicated mobile applications for students, supervisors, and internship providers to improve accessibility and convenience.

- **Automated Notifications:**

Implementing email and system notifications for report deadlines, evaluation submissions, and important internship updates.

- **Performance Analytics Dashboard:**

Developing analytical dashboards for administrators and supervisors to monitor student performance trends and internship progress.

- **Scalability Improvements:**

Optimizing the system architecture to handle larger numbers of students, reports, and concurrent users efficiently.

- **User Feedback Integration:**

Continuously collecting stakeholder feedback and applying iterative improvements to enhance system usability and functionality.

These future enhancements will make the Internship Management System more intelligent, scalable, and efficient, ensuring its long-term relevance and effectiveness in managing academic internships.

8 References

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